Coordinate System & Eigenvalues/Eigenvectors

Hung-yi Lee

Outline

- Each coordinate system is a "*viewpoint*" for vector representation.
  - The same vector/function is represented differently in different coordinate systems.
- Eigenvectors form a coordinate system to represent a function in a simply way (related to eigenvalues).

Coordinate System

A vector set $\mathcal{B}$ can be considered as a coordinate system on subspace V if $\mathcal{B}$ is a basis of V.

Coordinate System

Why basis?

- Let vector set $\mathcal{B}$=$\{u_1, u_2, \cdots, u_n\}$ be a basis for a subspace V

$\Rightarrow$ $\mathcal{B}$ is a coordinate system on V

- For any $\boldsymbol{v}$ in V, there are unique scalars $c_1, c_2, \cdots, c_n$ such that $\boldsymbol{v} = c_1\boldsymbol{u_1} + c_2\boldsymbol{u_2} + \cdots + c_n\boldsymbol{u_n}$

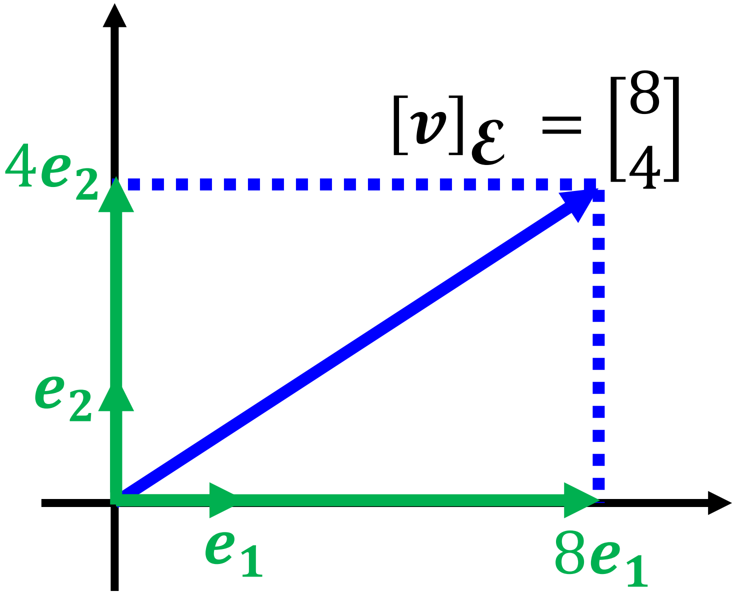
$\mathcal{B}$ -coordinate vector of $\boldsymbol{v}$:

$$[\boldsymbol{v}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

(用 $\mathcal{B}$ 的觀點來看 $\boldsymbol{v}$)

$\mathcal{E}$

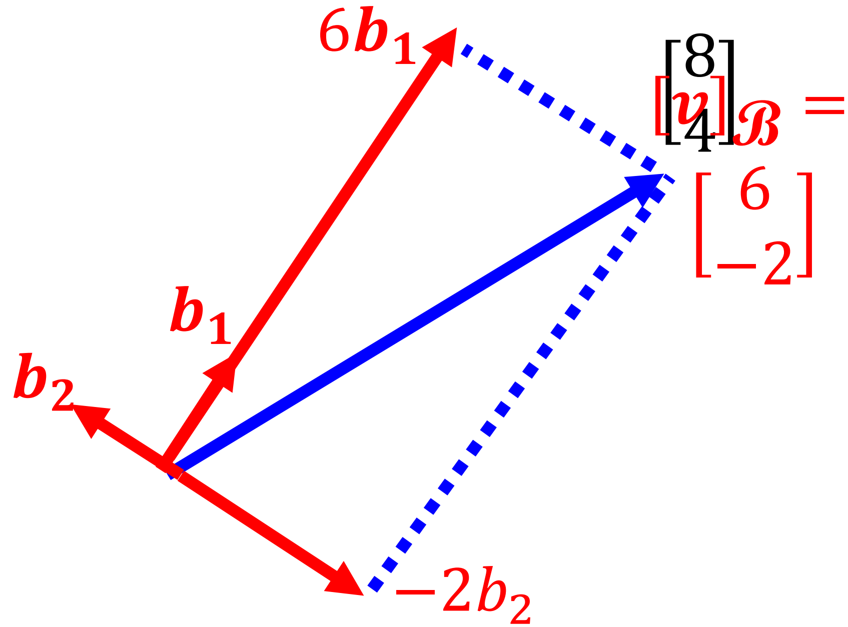
{$\mathbf{e}_1$, $\mathbf{e}_2$} is a coordinate system of R^2



$$\begin{bmatrix} 8 \\ 4 \end{bmatrix} = 8\boldsymbol{e}_1 + 4\boldsymbol{e}_2$$

New Coordinate System $\mathcal{B}$

$$\boldsymbol{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \boldsymbol{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$



$$\begin{bmatrix} 8 \\ 4 \end{bmatrix} = 6\boldsymbol{b}_1 + (-2)\boldsymbol{b}_2$$

$\mathcal{E}$={$\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n$}(standard vectors) $\qquad \boldsymbol{v} = [\boldsymbol{v}]_{\mathcal{E}}$

$\mathcal{E}$ is Cartesian coordinate system (直角坐標系)

Other System → Cartesian

- Let vector set $\boldsymbol{\mathcal{B}}$=$\{\boldsymbol{u_1}, \boldsymbol{u_2}, \cdots, \boldsymbol{u_n}\}$ be a coordinate system on a subspace R^n
- Matrix $\mathrm{B} = [\boldsymbol{u_1} \quad \boldsymbol{u_2} \quad \cdots \quad \boldsymbol{u_n}]$

Given $[\boldsymbol{v}]_{\boldsymbol{\mathcal{B}}}$, how to find $\boldsymbol{v}$? $\quad [\boldsymbol{v}]_{\boldsymbol{\mathcal{B}}} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$

$$\boldsymbol{v} = c_1\boldsymbol{u_1} + c_2\boldsymbol{u_2} + \cdots + c_n\boldsymbol{u_n}$$

$$= B[\boldsymbol{v}]_{\boldsymbol{\mathcal{B}}}$$ (matrix-vector product)

Cartesian ↔ Other System

- Let vector set $\mathcal{B}$=$\{\boldsymbol{u_1}, \boldsymbol{u_2}, \cdots, \boldsymbol{u_n}\}$ be a coordinate system on a subspace R^n

$$\boldsymbol{v} \underset{\boldsymbol{v} = B[\boldsymbol{v}]_{\mathcal{B}}}{\overset{[\boldsymbol{v}]_{\mathcal{B}} = B^{-1}\boldsymbol{v}}{\rightleftarrows}} [\boldsymbol{v}]_{\mathcal{B}}$$

$$= c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \cdots + c_n\mathbf{b}_n$$

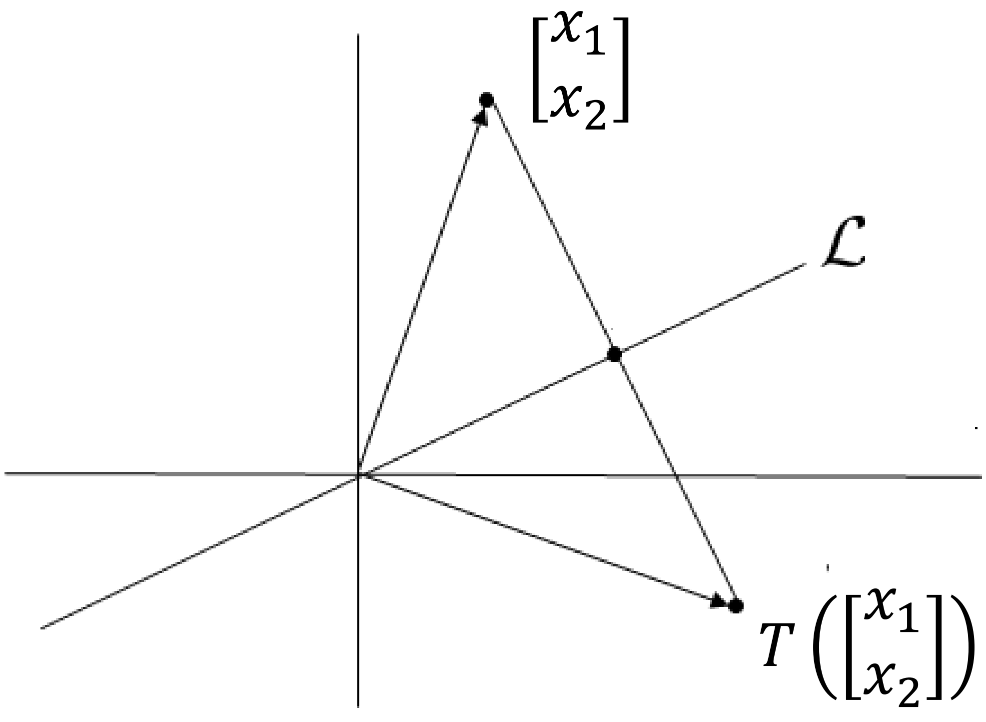
$$= \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

Let $\mathcal{B}$=$\{\boldsymbol{b_1}, \boldsymbol{b_2}, \cdots, \boldsymbol{b_n}\}$ be a basis of R^n. $[\boldsymbol{b_i}]_{\mathcal{B}} =? \boldsymbol{e_i}$

(Standard vector)

Changing Coordinate

- T: reflection about a line $\mathcal{L}$ through the origin in $\mathcal{R}^2$



$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = ?$$

$$[\mathrm{T}] = [T(e_1) \quad T(e_2)]$$

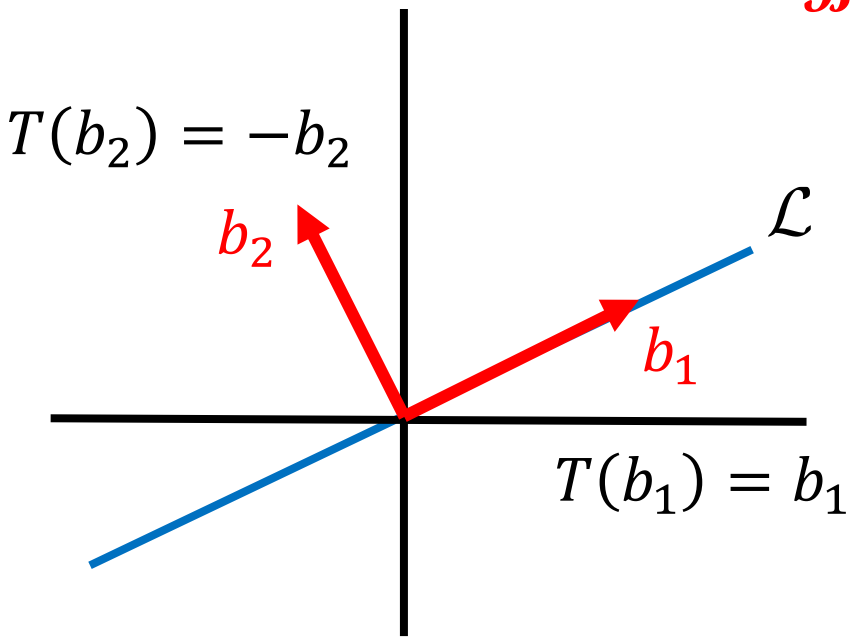
$$= ?$$

Changing Coordinate

- T: reflection about a line $\mathcal{L}$ through the origin in $\mathcal{R}^2$

$\mathcal{B} = \{b_1, b_2\}$

$\mathcal{B}$ is a coordinate system



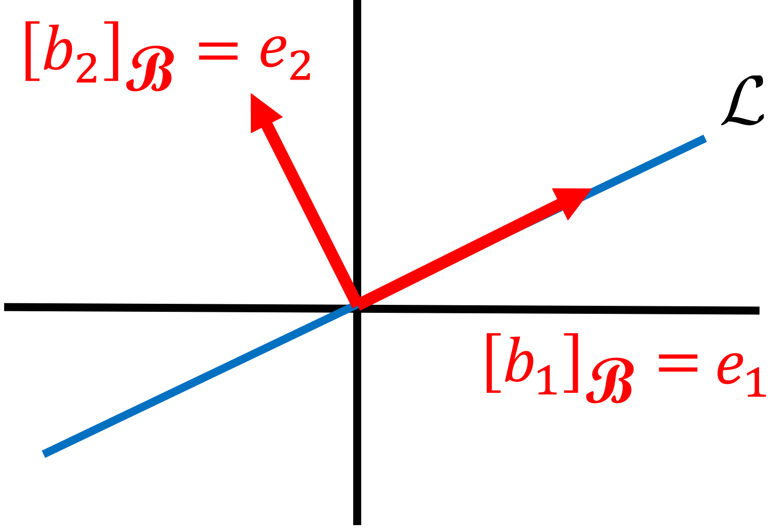
Changing Coordinate

- T: reflection about a line $\mathcal{L}$ through the origin in $\mathcal{R}^2$

Given the coordinate system $\mathcal{B}$ ....

$$[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$\mathcal{B}$ matrix of T: Input and output are both in $\mathcal{B}$



$[T]b_1 = b_1$

$\Rightarrow$ $[T]_{\mathcal{B}}([b_1]_{\mathcal{B}}) = [b_1]_{\mathcal{B}}$

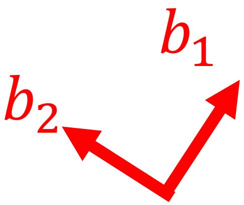
$\Rightarrow$ $[T]_{\mathcal{B}}(e_1) = e_1$

$[T]b_2 = -b_2$

$\Rightarrow$ $[T]_{\mathcal{B}}([b_2]_{\mathcal{B}}) = [-b_2]_{\mathcal{B}}$

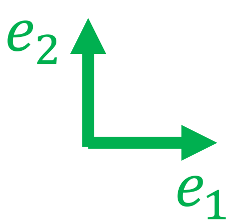
$\Rightarrow$ $[T]_{\mathcal{B}}(e_2) = -e_2$

Changing Coordinate



$$[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{tikzcd}
{[v]_{\mathcal{B}}} \arrow[rr, "\text{reflection about the horizontal line}"'] & & {[T(v)]_{\mathcal{B}}} \\
v \arrow[u] \arrow[rr, "{[T] = ?}", "\text{reflection about a line } \mathcal{L}"'] & & T(v) \arrow[u]
\end{tikzcd}$$



Flowchart

$\mathcal{B}$ coordinate system ?

$$
\begin{tikzcd}
{[v]_{\mathcal{B}}} \arrow[r, "{[T]_{\mathcal{B}}}"] & {[T(v)]_{\mathcal{B}}} \arrow[d, "B"] \\
v \arrow[u, "B^{-1}"] \arrow[r, "{[T]}"'] & T(v)
\end{tikzcd}
$$

Cartesian coordinate system

$$[T] = B[T]_{\mathcal{B}}B^{-1}$$

similar

$$[T]_{\mathcal{B}} = B^{-1}[T]B$$

similar

Eigenvalues and Eigenvectors

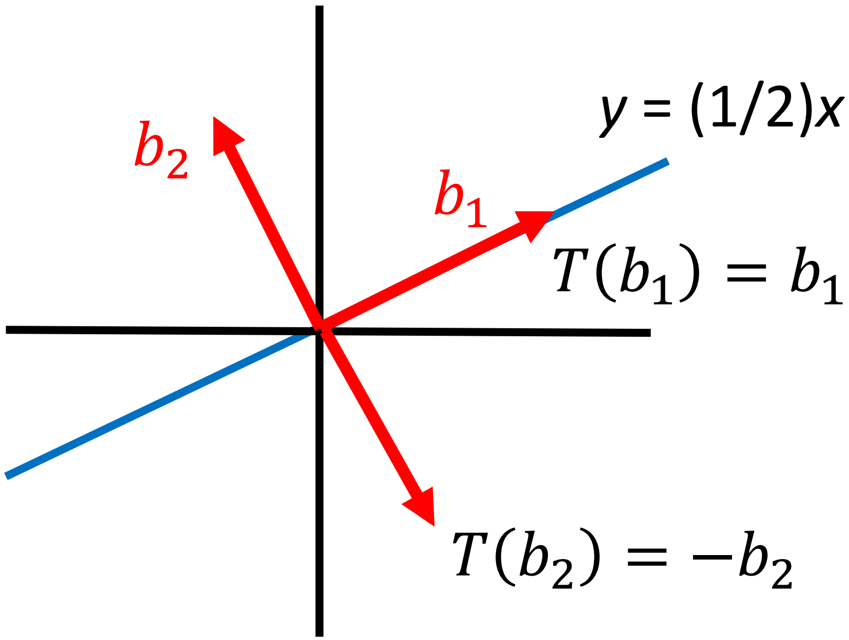
Eigenvalues and Eigenvectors

- If $A\boldsymbol{v} = \lambda\boldsymbol{v}$ ($\boldsymbol{v}$ is a vector, λ is a scalar)
  - $\boldsymbol{v}$ is an eigenvector of A excluding zero vector
  - λ is an eigenvalue of A that corresponds to $\boldsymbol{v}$
- T is a ***linear operator.*** If $\mathrm{T}(\boldsymbol{v}) = \lambda\boldsymbol{v}$ ($\boldsymbol{v}$ is a vector, λ is a scalar)
  - $\boldsymbol{v}$ is an eigenvector of T excluding zero vector
  - λ is an eigenvalue of T that corresponds to $\boldsymbol{v}$

Eigenvalues and Eigenvectors

- Example: Reflection

reflection operator *T* about the line *y* = (1/2)*x*



$\mathbf{b}_1$ is an eigenvector of *T*

Its eigenvalue is 1.

$\mathbf{b}_2$ is an eigenvector of *T*

Its eigenvalue is -1.

Eigenspace

- Assume we know λ is the eigenvalue of matrix A
- Eigenvectors corresponding to λ

$$A\mathbf{v} = \lambda\mathbf{v}$$

$$A\mathbf{v} - \lambda\mathbf{v} = \mathbf{0}$$

$$A\mathbf{v} - \lambda I_n\mathbf{v} = \mathbf{0}$$

$$\underbrace{(A - \lambda I_n)}_{\text{matrix}}\mathbf{v} = \mathbf{0}$$

Eigenvectors corresponding to λ are nonzero solution of

$$(A - \lambda I_n)\mathbf{v} = \mathbf{0}$$

Eigenvectors corresponding to λ

$$= \underbrace{Null(A - \lambda I_n)}_{\text{eigenspace}} - \{\mathbf{0}\}$$

Eigenspace of λ:

Eigenvectors corresponding to λ + $\{\mathbf{0}\}$

Check Eigenvalues

$Null(A - \lambda I_n)$:
eigenspace of λ

- How to know whether a scalar λ is the eigenvalue of A?

Check the dimension of eigenspace of λ

If the dimension is 0

⇒ Eigenspace only contains {0}

⇒ No eigenvector

⇒ λ is not eigenvalue

Looking for Eigenvalues

A scalar t is an eigenvalue of A

$\Leftrightarrow$ Existing $\boldsymbol{v} \neq 0$ such that $A\boldsymbol{v} = t\boldsymbol{v}$

$\Leftrightarrow$ Existing $\boldsymbol{v} \neq 0$ such that $A\boldsymbol{v} - t\boldsymbol{v} = 0$

$\Leftrightarrow$ Existing $\boldsymbol{v} \neq 0$ such that $(A - tI_n)\boldsymbol{v} = 0$

$\Leftrightarrow$ $(A - tI_n)\boldsymbol{v} = 0$ has multiple solution

$\Leftrightarrow$ The columns of $(A - tI_n)$ are Dependent

$\Leftrightarrow$ $(A - tI_n)$ is not invertible

$\Leftrightarrow$ $\boxed{det(A - tI_n) = 0}$

Characteristic Polynomial

$\boxed{\text{A scalar } t \text{ is an eigenvalue of A} \Longleftrightarrow det(A - tI_n) = 0}$

A is the standard matrix of linear operator T

$det(A - tI_n)$: Characteristic polynomial of A

linear operator T

$det(A - tI_n) = 0$: Characteristic equation of A

linear operator T

Eigenvalues are the roots of characteristic polynomial or solutions of characteristic equation.

Looking for Eigenvalues/Eigenspaces

- Example: Find the eigenvalues of $A = \begin{bmatrix} -4 & -3 \\ 3 & 6 \end{bmatrix}$

$$\boxed{\text{A scalar } t \text{ is an eigenvalue of A} \Leftrightarrow det(A - tI_n) = 0}$$

$$A - tI_2 = \begin{bmatrix} -4-t & -3 \\ 3 & 6-t \end{bmatrix}$$

$\det(A - tI_2)$ =0

$\Rightarrow$ t = -3 or 5

The eigenvalues of *A* are -3 or 5.

Looking for Eigenvalues/Eigenspaces

- Example: Find the eigenvalues of $A = \begin{bmatrix} -4 & -3 \\ 3 & 6 \end{bmatrix}$

The eigenvalues of *A* are -3 or 5.

<u>Eigenspace of -3</u>

$$Ax = -3x \Rightarrow (A + 3I)x = 0$$

find the solution

<u>Eigenspace of 5</u>

$$Ax = 5x \Rightarrow (A - 5I)x = 0$$

find the solution

Characteristic Polynomial

- In general, a matrix A and RREF of A have different characteristic polynomials. $\Rightarrow$ Different Eigenvalues
- Similar matrices have the same characteristic polynomials $\Rightarrow$ The same Eigenvalues

$$\boxed{B = P^{-1}AP}$$

$$\begin{aligned} det(B - tI) &= det(P^{-1}AP - P^{-1}(tI)P) \\ &= det(P^{-1}(A - tI)P) \\ &= det(P^{-1})det(A - tI)det(P) \\ &= \left(\frac{1}{det(P)}\right)det(A - tI)det(P) = det(A - tI) \end{aligned}$$

Characteristic Polynomial

- Question: What is the order of the characteristic polynomial of an $n \times n$ matrix A?
  - The characteristic polynomial of an $n \times n$ matrix is indeed a polynomial with degree n
  - Consider $\det(A - tI_n)$

$$det\begin{bmatrix} ?-t & \cdots & ? \\ \vdots & \ddots & \vdots \\ ? & \cdots & ?-t \end{bmatrix}$$

- Question: What is the number of eigenvalues of an $n \times n$ matrix A?
  - Fact: An n x n matrix A have less than or equal to n eigenvalues
  - Consider complex roots and multiple roots

Characteristic Polynomial vs. Eigenspace v.s. Eigenvectors

$det(A - tI_n)$ **Factorization**

$$= (t-\lambda_1)^{m_1}(t-\lambda_2)^{m_2} \dots (t-\lambda_k)^{m_k}(\dots\dots)$$

Eigenvalue: λ_1 λ_2 …… λ_k

Eigenspace: (dimension) $d_1 \leq m_1$ $d_2 \leq m_2$ …… $d_k \leq m_k$

Independent

A set of eigenvectors that correspond to distinct eigenvalues is linear independent.

Same Eigenvalues

(b) Let A, B and P be $n \times n$ matrices. Suppose that $A = P^{-1}BP$.
Please answer whether A and B have the same eigenvalues or not. (3%)
Justify your answer. (3%)

(2014)

6. **(10%)** Let A and B be $n \times n$ matrices such that $A = P^{-1}BP$ for an invertible $n \times n$ matrix P. Please show that the characteristic polynomial of A is the same as that of B.

(2012)

---

2. (8%) Let A be an $n \times n$ matrix. Prove that $A^T A$ and AA^T have the same set of eigenvalues.

(2018)

$$A^T A v = \lambda v \Leftrightarrow AA^T u = \lambda u$$

$$\Rightarrow: \quad A^T A v = \lambda v \quad \Rightarrow AA^T A v = \lambda A v \quad \Rightarrow AA^T \underbrace{(Av)}_{u} = \lambda \underbrace{(Av)}_{u}$$

Find Eigenvalues in a fast way

3. Given a matrix

(2018)

$$A = \begin{bmatrix} 1 & 6 & -6 & -6 \\ 6 & 7 & -6 & -12 \\ 3 & 3 & -2 & -6 \\ 3 & 9 & -9 & -11 \end{bmatrix},$$

whose eigenvalues are -5, -2, and 1.

(a) (3%) Please find the eigenvalues of A^{-1}. 1/(-5), 1/(-2), 1

(b) (3%) Please find the eigenvalues of A^2.

(c) (6%) Please find the eigenvalues of $A^3 + 2A$.

---

9. (a) If λ_1 is one eigenvalue of a invertible real matrix A, the corresponding eigenvector is $\vec{\mathbf{p}}_1$. Show that $(\frac{1}{\lambda_1})$ is one eigenvalue of A^{-1}, the corresponding eigenvector is $\vec{\mathbf{p}}_1$. (10%)

(2013)

A is invertible = 0 is not eigenvalue (exercise 66. P301)

Find Eigenvalues in a fast way

3. Given a matrix

(2018)

$$A = \begin{bmatrix} 1 & 6 & -6 & -6 \\ 6 & 7 & -6 & -12 \\ 3 & 3 & -2 & -6 \\ 3 & 9 & -9 & -11 \end{bmatrix},$$

whose eigenvalues are -5, -2, and 1.

(a) (3%) Please find the eigenvalues of A^{-1}. 1/(-5), 1/(-2), 1

(b) (3%) Please find the eigenvalues of A^2. $(-5)^2$, $(-2)^2$, 1

(c) (6%) Please find the eigenvalues of $A^3 + 2A$.

$(-5)^3+2(-5)$, $(-2)^3+2(-2)$, $(1)^3+2(1)$

A	$a_n A^n + a_{n-1} A^{n-1} + \dots\dots + a_1 A + a_0 I$
λ	$a_n \lambda^n + a_{n-1} \lambda^{n-1} + \dots\dots + a_1 \lambda + a_0$

5. (8%) A square matrix A is a nilpotent matrix if there exists a positive integer p such that $A^p = O$. Show that the only eigenvalue of A is 0.

(2019)

If λ is A's eigenvalue $\mathrm{A}\boldsymbol{v} = \lambda\boldsymbol{v}$ $\boldsymbol{v} \neq \mathbf{0}$

$$\Rightarrow \mathrm{AA}\boldsymbol{v} = \lambda\mathrm{A}\boldsymbol{v} \Rightarrow \mathrm{A}^2\boldsymbol{v} = \lambda^2\boldsymbol{v}$$

$$\mathrm{A}^p\boldsymbol{v} = \lambda^p\boldsymbol{v} \qquad \lambda^p = 0 \qquad \lambda = 0$$

$$= \mathbf{0}$$

4. Find *all* the possible eigenvalues that the $n \times n$ matrix A satisfying the following condition *can* have.

(a) (6%) $A^2 = A$.

(b) (4%) $A^k = O$ for some positive integer k

(2020)

If λ is A's eigenvalue $\mathrm{A}\boldsymbol{v} = \lambda\boldsymbol{v}$ $\boldsymbol{v} \neq \mathbf{0}$

$$\Rightarrow \mathrm{AA}\boldsymbol{v} = \lambda\mathrm{A}\boldsymbol{v} \Rightarrow \mathrm{A}^2\boldsymbol{v} = \lambda^2\boldsymbol{v} \Rightarrow A\boldsymbol{v} = \lambda^2\boldsymbol{v}$$

$$\lambda = \lambda^2 \qquad \lambda = 1 \; or \; 0$$

Problem 3 (11%)
Suppose that A is an n by n matrix and the characteristic polynomial of A is given by $-t^5+2$.

(a) (1%+2%+2%) Find the values of n, det A, and trace A.

(2021)

(b) (6%) Find the characteristic polynomial of A^2.

(a) $n = 5$

$t = 0$

$det(A - tI_n) = -t^5 + 2$ $\quad$ $det(A) = 2$

$traceA = 0$ Check the coefficient of t^4

coefficient of t^4 = $traceA$

(1:06:00)

https://drive.google.com/file/d/1UK3nfOyEK9U4rGKZvoyxBwr4v3lBrQX_/view

(b) $det(A^2 - tI_n) = det\left(\left(A - \sqrt{t}I_n\right)\left(A + \sqrt{t}I_n\right)\right)$

$= det\left(A - \sqrt{t}I_n\right)det\left(A + \sqrt{t}I_n\right) = -t^5 + 4$

$-t^{2.5} + 2$ $\quad$ $t^{2.5} + 2$

Problem 4 (12%)
Let

$$A = \begin{bmatrix} -5 & 6 & 1 \\ -1 & 2 & 1 \\ -8 & 6 & 4 \end{bmatrix},$$

(2021)

whose eigenvalues are 2, 3 and -4.

(a) (6%) Find the eigenvalues of $\left(A^2 + 2A\right)^T$

(b) (6%) The eigenvectors that we have been studying in this course is sometimes called right eigenvectors to distinguish them from left eigenvectors, which are vectors $\boldsymbol{x}$ satisfy the equation $\boldsymbol{x}^T A = \mu \boldsymbol{x}^T$ for some scalar μ. Find the left eigenvalues of A.

$$det(A - tI) = det\left((A - tI)^{\mathrm{T}}\right) = det\left(\mathrm{A}^{\mathrm{T}} - tI^T\right) = det(A^T - tI)$$

(a) Simply find the eigenvalues of $\mathrm{A}^2 + 2A$

(b) $\boldsymbol{x}^T A = \mu \boldsymbol{x}^T$ $\qquad A^T \boldsymbol{x} = \mu \boldsymbol{x}$

A's left eigenvalues = A^T's (right) eigenvalues

Diagonalization

An nxn matrix A is called diagonalizable if $A = PDP^{-1}$

D: nxn diagonal matrix

P: nxn invertible matrix

Not all matrices are diagonalizable

Diagonalization of Linear Operator

If a linear operator T is diagonalizable

$$\begin{tikzcd}[column sep=huge, row sep=huge]
{[\boldsymbol{v}]_{\mathcal{B}}} \arrow[r, "D", "\text{simple}"'] & {[T(\boldsymbol{v})]_{\mathcal{B}}} \arrow[d, "P", "\text{Properly selected}"'] \\
\boldsymbol{v} \arrow[u, "P^{-1}", "\text{Properly selected}"'] \arrow[r, "A = PDP^{-1}"] & T(\boldsymbol{v})
\end{tikzcd}$$

Diagonalizable

$$P = [\boldsymbol{p_1} \quad \cdots \quad \boldsymbol{p_n}]$$

$$D = \begin{bmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_n \end{bmatrix}$$

$$= [d_1\boldsymbol{e_1} \quad \cdots \quad d_n\boldsymbol{e_n}]$$

- If A is diagonalizable

$$A = PDP^{-1} \Rightarrow AP = PD$$

$$AP = [\underline{A\boldsymbol{p_1}} \quad \cdots \quad \underline{A\boldsymbol{p_n}}]$$

$$PD = P[d_1\boldsymbol{e_1} \quad \cdots \quad d_n\boldsymbol{e_n}]$$

$$= [Pd_1\boldsymbol{e_1} \quad \cdots \quad Pd_n\boldsymbol{e_n}]$$

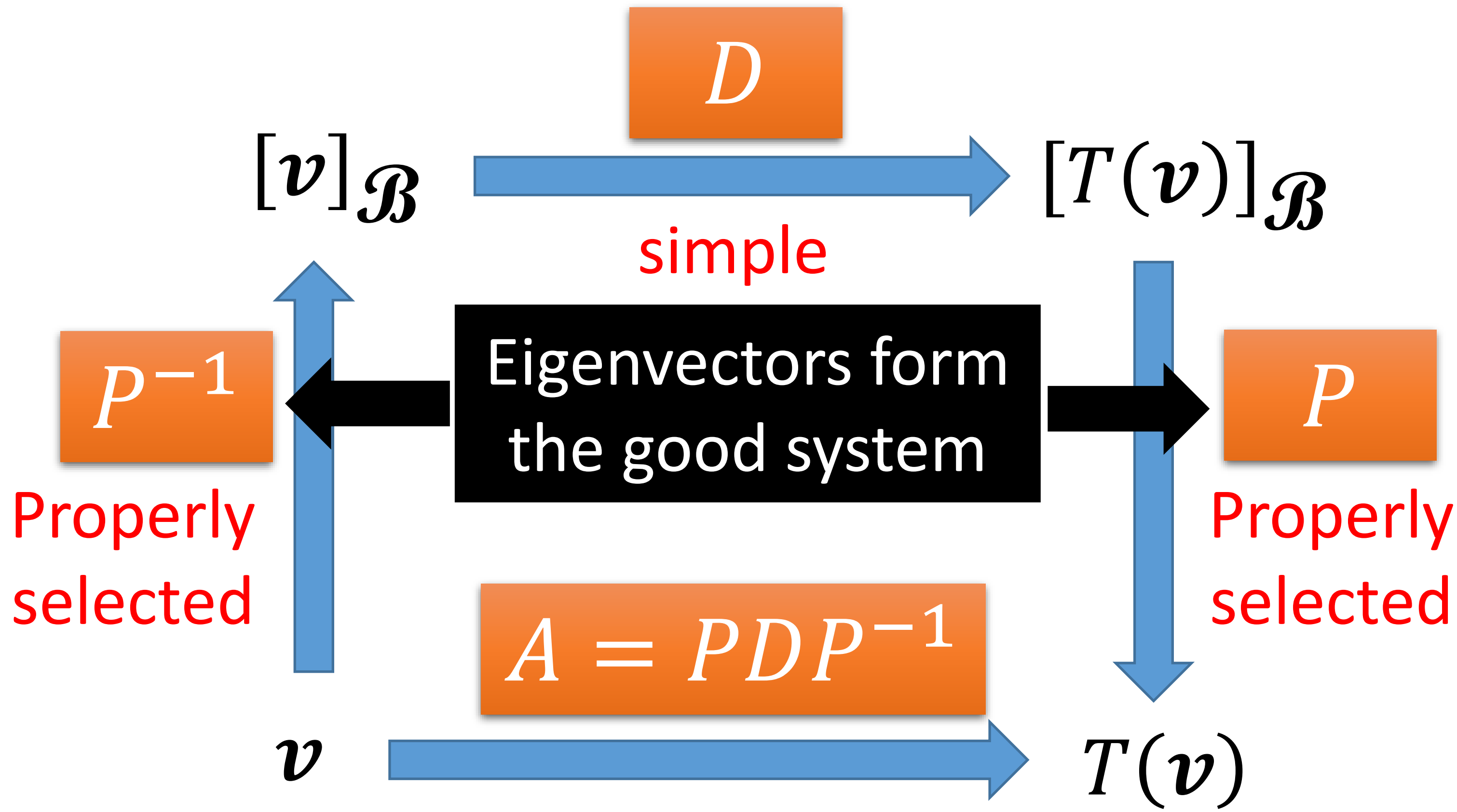
$$= [d_1P\boldsymbol{e_1} \quad \cdots \quad d_nP\boldsymbol{e_n}]$$

$$= [\underline{d_1\boldsymbol{p_1}} \quad \cdots \quad \underline{d_n\boldsymbol{p_n}}] \Rightarrow A\boldsymbol{p_i} = d_i\boldsymbol{p_i}$$

$\boldsymbol{p_i}$ is an eigenvector of A corresponding to eigenvalue d_i

Diagonalization of Linear Operator

- If a linear operator T is diagonalizable



$$P = [\boldsymbol{p}_1 \quad \cdots \quad \boldsymbol{p}_n]$$

Diagonalizable

$$D = \begin{bmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_n \end{bmatrix}$$

- If A is diagonalizable

$$A = PDP^{-1}$$

$\boldsymbol{p}_i$ is an eigenvector of A corresponding to eigenvalue d_i

$\parallel$

There are n eigenvectors that form an invertible matrix

$\parallel$

There are n independent eigenvectors

$\parallel$

The eigenvectors of A can form a basis for R^n.

Diagonalizable

$$P = [\boldsymbol{p}_1 \quad \cdots \quad \boldsymbol{p}_n]$$

$$D = \begin{bmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_n \end{bmatrix}$$

- If A is diagonalizable

$$A = PDP^{-1}$$

$\boldsymbol{p}_i$ is an eigenvector of A corresponding to eigenvalue d_i

How to diagonalize a matrix A?

Step 1: Find *n* independent eigenvectors corresponding if possible, and form an invertible *P*

Step 2: The eigenvalues corresponding to the eigenvectors in P form the diagonal matrix *D.*

Diagonalizable

$$P = [\boldsymbol{p}_1 \quad \cdots \quad \boldsymbol{p}_n]$$

$$D = \begin{bmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_n \end{bmatrix}$$

- If A is diagonalizable

$$A = PDP^{-1}$$

$\boldsymbol{p}_i$ is an eigenvector of A corresponding to eigenvalue d_i

$$det(A - tI_n)$$
$$= (t-\lambda_1)^{m_1}(t-\lambda_2)^{m_2} \ldots (t-\lambda_k)^{m_k}(\ldots \ldots)$$

Eigenvalue: λ_1 λ_2 ...... λ_k

Eigenspace: ......

$\{$ Basis for λ_1 Basis for λ_2 Basis for λ_3 $\}$

Independent Eigenvectors

You can't find more!

Diagonalizable - Example

- Diagonalize a given matrix $A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix}$

characteristic polynomial is $-(t+1)^2(t-3)$ $\Rightarrow$ eigenvalues: 3, -1

eigenvalue 3

$$\mathbb{B}_1 = \left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

eigenvalue -1

$$\mathbb{B}_2 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$$

$A = PDP^{-1}$, where

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & -1 \end{bmatrix}$$

$$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

3. (10%) Determine all values of c such that the following matrix is not diagonalizable.

$$\begin{bmatrix} 1 & 0 & -1 \\ -2 & c & -2 \\ 2 & 0 & 4 \end{bmatrix} \qquad \text{(2021)}$$

An n x n matrix A is diagonalizable

=

The eigenvectors of A can form a basis for R^n.

Eigenvalue: λ_1 λ_2 …… λ_k

Eigenspace: d_1 d_2 …… d_k

(dimension)

$$d_1 + d_2 + \ldots + d_k = n$$

3. (10%) Determine all values of c such that the following matrix is not diagonalizable.

$$A = \begin{bmatrix} 1 & 0 & -1 \\ -2 & c & -2 \\ 2 & 0 & 4 \end{bmatrix}$$ (2011)

Characteristic polynomial of A:

$$(1-t)(c-t)(4-t)+2(c-t)$$
$$=(c-t)[(1-t)(4-t)+2]$$
$$=(c-t)(6-5t+t^2)=(c-t)(3-t)(2-t)$$

Check ...

If $c \neq 2$ and $c \neq 3$

A is diagonalizable

$$\begin{bmatrix} 1 & 0 & -1 \\ -2 & 2 & -2 \\ 2 & 0 & 4 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & -1 \\ -2 & 3 & -2 \\ 2 & 0 & 4 \end{bmatrix}$$

Diagonalization of Linear Operator

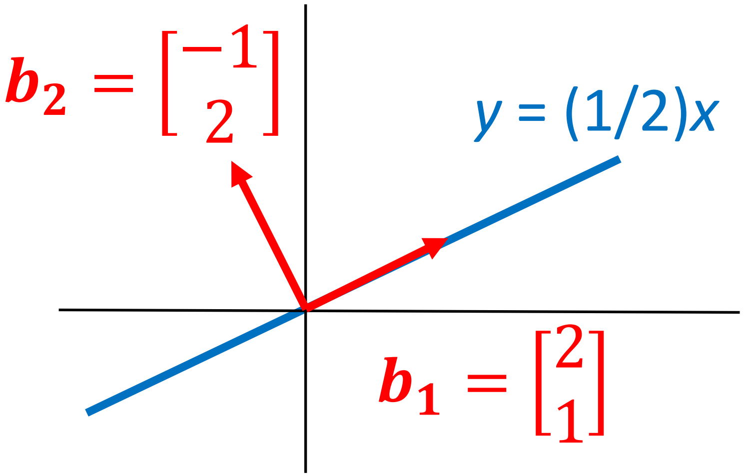
$$T\left(\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}\right)=\begin{bmatrix}8x_1+9x_2\\-6x_1-7x_2\\3x_1+3x_2-x_3\end{bmatrix}$$

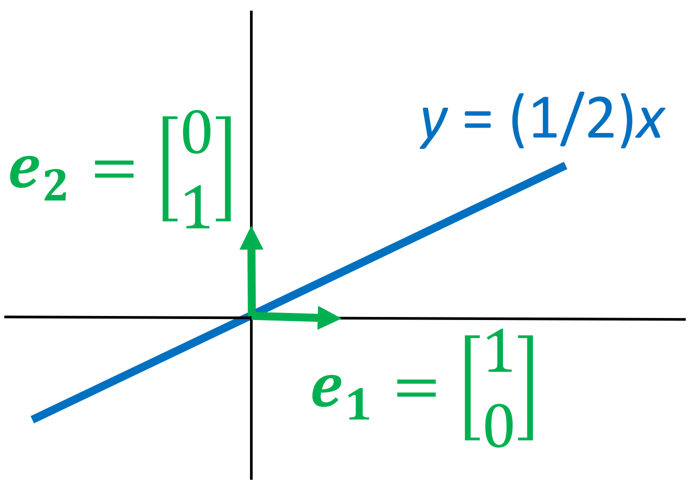
-1: $\mathcal{B}_1=\left\{\begin{bmatrix}-1\\1\\0\end{bmatrix},\begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$

2: $\mathcal{B}_2=\left\{\begin{bmatrix}3\\-2\\1\end{bmatrix}\right\}$

$$\begin{tikzcd}[column sep=huge, row sep=huge]
{[v]_{\mathcal{B}}} \arrow[r, "{[T]_{\mathcal{B}}}", "{\begin{bmatrix}-1&0&0\\0&-1&0\\0&0&2\end{bmatrix}}"'] & {[T(v)]_{\mathcal{B}}} \arrow[d, "{\begin{bmatrix}-1&0&3\\1&0&-2\\0&1&1\end{bmatrix}}"] \\
v \arrow[u, "{\begin{bmatrix}-1&0&3\\1&0&-2\\0&1&1\end{bmatrix}^{-1}}"] \arrow[r, "{\begin{bmatrix}8&9&0\\-6&-7&0\\3&3&-1\end{bmatrix}}"] & T(v)
\end{tikzcd}$$

- Example: reflection operator T about the line $y = (1/2)x$





$$B^{-1} = \begin{bmatrix} 0.4 & 0.2 \\ -0.2 & 0.4 \end{bmatrix} \quad B = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$$

$$\begin{tikzcd}
{[v]_{\mathcal{B}}} \arrow[r, "{[T]_{\mathcal{B}}}"] & {[T(v)]_{\mathcal{B}}} \arrow[d, "B"] \\
v \arrow[u, "B^{-1}"] \arrow[r, "{[T] =?}"] & T(v)
\end{tikzcd}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$[T] = B[T]_{\mathcal{B}}B^{-1}$$

Diagonalization of Linear Operator

- If a linear operator T is diagonalizable